Aarav Verma

Aerospace Engineering Student — University of Illinois Urbana-Champaign

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Aerospace Engineering student at UIUC. I build hardware that flies and moves — currently leading drone development at NOBE, engineering drivetrain systems at Ghost Electric Motorcycles, and founding Volant, a fleet operations platform for drone and eVTOL operators.

EDUCATION

B.S. Aerospace Engineering, CS Minor — UIUC (2024–2028)

CURRENT FOCUS

eVTOL fleet management software and autonomous UAV systems

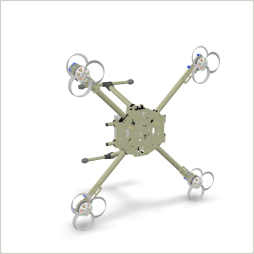
ORGANIZATIONS

NOBE — Project Manager

Ghost Electric Motorcycles — Drivetrain Engineer

Illinois Space Society — Space Shot Structural Engineer

CAD & HARDWARE PROJECTS



Full Drone + Computer System

Feb 2026–Present

NOBE — Engineering Lead

Designed a modular drone airframe in Fusion with a Jetson housing and foldable leg motors on 4 snap-lock hinges — no tools needed to swap configs in the field. FEA under 12N thrust load confirmed FOS above 2.0. BOM covers 50+ components.

UAV | Avionics | Fusion 360 | FEA



Toroidal Propeller Design

Feb 2026–Present

NOBE — Engineering Lead

Redesigned a 2-blade propeller to toroidal config in Fusion — enclosing the blade-tip vortex path pushed thrust efficiency up ~11% at 4–6k RPM and dropped acoustic signature below 60 dB at 1m hover. FEA on the propeller and motor mount under 12N load confirmed FOS above 2.0.

Propulsion | Aerodynamics | Fusion 360 | FEA



Swingarm Assembly

Aug 2025–Present

Ghost Electric Motorcycles — Drivetrain Engineer

Rear suspension swingarm in SolidWorks, reverse-engineered from the 2010 Kawasaki ZX-6R. Worked out anti-squat ratio and pivot geometry to match OEM kinematics, then validated stiffness under braking, acceleration, and cornering in ANSYS and MATLAB.

SolidWorks | FEA | Suspension | ANSYS



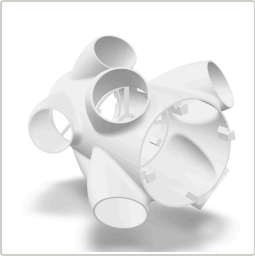
GHOST — Drivetrain Frame

Aug 2025–Present

Ghost Electric Motorcycles — Drivetrain Engineer

Drivetrain frame in SolidWorks to mount and align motor, top rod, and swingarm to ±0.05 mm. Structural analysis in ANSYS and MATLAB confirmed stiffness under peak torque, chain tension, and rider weight across braking, cornering, and full acceleration.

SolidWorks | FEA | ANSYS



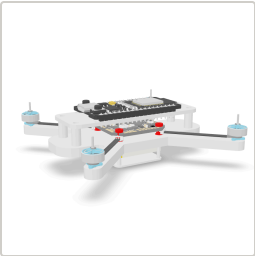
Revolving Modular Hydroponic Tower

2024

UIUC Residence Hall Idea Fair — Designer & Builder

Revolving modular hydroponic tower designed in Fusion. Built the stacking system around FDM print constraints — tolerances defined for every interlocking joint. Printed and assembled the full prototype in-house.

Fusion 360 | 3D Printing | DFM



Volant Demo Drone

2025–Present

Volant (Personal Startup) — Founder & Designer

Palm-sized drone airframe designed in Fusion as the hardware side of Volant, my fleet ops platform for drone and eVTOL operators. Modeled to FDM tolerances so the whole thing is printable and field-deployable, built to run in a 10-drone swarm tied into Volant's telemetry and mission dispatch.

Fusion 360 | 3D Printing | UAV

SIMULATION & ANALYSIS — MATLAB / SIMULINK

Cascaded Flight Controller — Pitch/Roll/Yaw

2026

Simulink · Controls

Built a multi-axis PID attitude-control system in Simulink from first principles — derived plant transfer functions from Newton's second law, tuned via manual and Ziegler–Nichols methods, then restructured pitch into a cascaded inner-rate/outer-angle architecture matching real flight-controller designs (Betaflight, PX4). Diagnosed and fixed derivative-kick and noise-amplification issues by isolating root cause rather than patching with filters, and validated disturbance rejection against injected wind-gust and sensor-noise models.

Simulink | PID Control | MATLAB

CAD-to-FEA Pipeline — NACA 0012 Airfoil

2026

MATLAB PDE Toolbox · FEA

Generated NACA 0012 airfoil geometry programmatically in MATLAB, built the 3D solid in Fusion 360, and imported it into MATLAB's Structural Mechanics FEA solver for stress/displacement analysis under applied load. Ran a formal mesh convergence study, caught a stress singularity at an unfilleted corner, and corrected it through CAD redesign — then re-validated smooth convergence on the fixed geometry.

MATLAB | FEA | Mesh Convergence

Cantilever Bracket — Hand-Validated FEA

2026

MATLAB PDE Toolbox · FEA

Designed a dimensioned sheet-metal cantilever bracket and independently hand-calculated tip deflection and bending stress before simulating, then diagnosed a 10x discrepancy between hand calc and FEA by tracing it to an incorrect assumed load-face area — fixed by querying the true face area directly from the solved mesh. Reconciled the corrected FEA output against hand theory, attributing the remaining gap to bend-fillet compliance and hole stress concentration.

MATLAB | FEA | Mechanics of Materials

SOFTWARE PROJECTS

Volant

Startup · B2B SaaS — github.com/Verma0118/volant

Fleet operations platform for drone and eVTOL operators — real-time telemetry, mission dispatch, and fleet management.

React | TypeScript | Node.js | PostgreSQL

TVC Simulator

Simulation Tool — github.com/Verma0118/TVC_Simulator_V1

Thrust Vector Control simulator built in Python. Models gimbal-actuated rocket engine dynamics for attitude control during powered flight.

Python | Simulation | Controls

SKILLS

CAD & SIMULATION

NX Siemens, SolidWorks, CATIA V5, ANSYS, MATLAB, Simulink

SOFTWARE

Python, TypeScript, React, C++, Git, ROS

MANUFACTURING

FDM 3D Printing, CNC Machining, Laser Cutting, Sheet Metal Fabrication

COURSEWORK — AE 140

Introduction to CAD for Aerospace Engineers. 10 modeled assemblies including a UAV design, spur gear, front sprocket, and multiple tutorial/homework exercises in NX.
